Numerical Analysis (089180) A.Y. 2024-2025
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Lab 1 – Homework Solutions

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A Homework

Homework 1.1 (Error propagation). The sequence

$$1, \frac{1}{3}, \frac{1}{9}, \ldots, \frac{1}{3^n}, \ldots$$

can be generated with the following recursive relations

$$\begin{cases} p_n = \dfrac{10}{3} p_{n-1} - p_{n-2} \\ p_1 = \frac{1}{3}, p_0 = 1 \end{cases} \qquad \begin{cases} q_n = \dfrac{1}{3} q_{n-1} \\ q_0 = 1. \end{cases}$$

a. Implement the two relations in order to generate the first 100 terms of the sequence.

b. Study the stability of the two algorithms and justify the obtained results.

Solution Homework 1.1.

a. Implement the two relations in order to generate the first 100 terms of the sequence.

```
% First recursive relation
p(1) = 1;
p(2) = 1/3;
for i = 2:100
   p(i+1) = 10/3*p(i) - p(i-1);
end

figure
subplot(2,1,1)
plot(0:100, p, 'LineWidth',3)
set(gca,'FontSize',16)
xlabel('n','FontSize',16)
ylabel('p_n','FontSize',16)
% The sequence explodes!

% Second recursive relation
q(1) = 1;
for i=1:100
   q(i+1) = 1/3*q(i);
end

subplot(2,1,2)
plot(0:100, q, 'LineWidth',3)
set(gca,'FontSize',16)
xlabel('n','FontSize',16)
ylabel('q_n','FontSize',16)

% The sequence instead is ok.
```

b. Study the stability of the two algorithms and justify the obtained results.

Error analysis: since we use finite precision numbers, we know the initial data up to an error ε[1]Indeed, denoting by $fl(p_i)$ the floating point representative for p_i

$$fl(p_0) = p_0 + \varepsilon, \qquad fl(p_1) = p_1 + \varepsilon.$$

It follows

$$\begin{aligned} fl(p_2) &= \frac{10}{3} fl(p_1) - fl(p_0) = \frac{10}{3}(p_1 + \varepsilon) - (p_0 + \varepsilon) = p_2 + \frac{7}{3}\varepsilon \\ fl(p_3) &= \frac{10}{3} fl(p_2) - fl(p_1) = \frac{10}{3}(p_2 + \frac{7}{3}\varepsilon) - (p_1 + \varepsilon) = p_2 + \frac{61}{9}\varepsilon \\ \tilde{p}_4 &= ... \end{aligned}$$

The error is amplified!Instead, with the second formula

$$\begin{aligned} fl(q_0) &= q_0 + \varepsilon \\ fl(q_1) &= \frac{1}{3} fl(q_0) = \frac{1}{3} q_0 + \frac{1}{3}\varepsilon = q_1 + \frac{1}{3}\varepsilon \\ fl(q_2) &= \frac{1}{3} fl(q_1) = \frac{1}{3} q_1 + \frac{1}{9}\varepsilon = q_2 + \frac{1}{9}\varepsilon \\ fl(q_3) &= ... \end{aligned}$$

The error is reduced!

Homework 1.2. Find the minimum positive number representable in MATLAB by implementing an ad hoc procedure. Compare with `realmin`.

Solution Homework 1.2.

```
k = 0;
zero = 1/2;
while (0 + zero) > 0
  zero_old = zero;
  zero = zero / 2;
  k = k + 1;
end

% Remember: realmin is the minimum !normalized! positive floating point number
realmin
% The minimum positive floating point number is instead a !denormalized! number
zero_old % < realmin
k
```

Homework 1.3. a. Use Taylor polynomial approximation to avoid the loss of significance errors in the following function when x approaches 0

$$f(x) = \frac{1 - \cos(x)}{x^2}$$

b. Reformulate the following function $g(x)$ to avoid the loss of significance error in its evaluation for increasing values of x towards $+\infty$

$$g(x) = x\left(\sqrt{x+1} - \sqrt{x}\right).$$

Solution Homework 1.3.

```
format long
```

[1] Actually, if we start from 1, we kno it exactly, but we may introduce anyway errors in the following steps.

a. Use Taylor polynomial approximation to avoid the loss of significance errors in the following function when x approaches 0

$$f(x) \;=\; \frac{1-\cos(x)}{x^2}$$

The limit of $f(x) = \dfrac{1-\cos(x)}{x^2}$ as $x \to 0$ is well known:

$$\lim_{x\to 0} \frac{1-\cos(x)}{x^2} = \frac{1}{2}$$

```
clear all, close all, c
k = [1:30]';
x = 2.^(-k);
f = @(x) (1 - cos(x))./(x.^2);
[k x f(x)]
```

If we try to evaluate $f(x)$ directly with a sequence $\{x_k = 2^{-k}\}_k$ that approaches 0, we notice anamalous behaviour at $k = 13$, $k = 27$, $k = ...$The reason is numerical cancellation of the terms on the numerator. We can use Taylor expansion of $\cos(x)$ for $x \to 0$ to obtain a better approximation. Indeed

$$\cos(x) = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6 + o(x^8)$$

so that

$$f(x) = \frac{1}{2} - \frac{1}{24}x^2 + \frac{1}{720}x^4 + o(x^6)$$

```
f_taylor_4 = @(x) 1/2 - x.^2/24 + x.^4/720;
```

```
[k x f(x) f_taylor_4(x)]
```

The obtained formula is more stable that the direct evaluation of f(x), and the computed values approach $\frac{1}{2}$.

b. Reformulate the following function $g(x)$ to avoid the loss of significance error in its evaluation for increasing values of x towards $+\infty$

$$g(x) \;=\; x\Big(\sqrt{x+1} - \sqrt{x}\Big).$$

It is well known that

$$\lim_{x\to +\infty} g(x) = +\infty$$

```
clear all, close all, clc
format short e

k = [1:20]';
x = 10.^(k);

g = @(x) x.*(sqrt(x+1) - sqrt(x));

[k x g(x)]
```

If we try to evaluate $g(x)$ for a sequence $\{x_k = 10^k\}_k$ that approaches 0, we notice anamalous behaviour at $k = 16$, $k = ...$, for which the computed value is 0. This happens since, in floating point representation, $x_{16} + 1 = x_{16}$!Rationalization of the expression in bracket solves this numerical cancellation: indeed, multiplying the numerator and the denominator by $\sqrt{x+1} + \sqrt{x}$

$$g(x) = \frac{x\left(\sqrt{x+1} - \sqrt{x}\right)\left(\sqrt{x+1} + \sqrt{x}\right)}{\left(\sqrt{x+1} + \sqrt{x}\right)} = \frac{x}{\left(\sqrt{x+1} + \sqrt{x}\right)}.$$

```
g2 = @(x) x./(sqrt(x+1) + sqrt(x));

[k x g(x) g2(x)]
```

This formula is more stable and does not suffer of numerical cancellation.

Homework 1.4. We can compute e^{-x} around $x = 0$ using Taylor polynomials in two ways, either using

$$e^{-x} \approx 1 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \cdots$$

or using

$$e^{-x} = \frac{1}{e^x} \approx \frac{1}{1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots}.$$

Which approach is the most accurate?

Solution Homework 1.4.

```
clear all, close all, clc
format long

f_taylor_pos = @(x) 1 - x + 1/2*x.^2 - 1/6*x.^3 + 1/24*x.^4 - 1/120*x.^5;
f_taylor_neg = @(x) 1./(1 + x + 1/2*x.^2 + 1/6*x.^3 + 1/24*x.^4 + 1/120*x.^5);

k = [1:20]';

x_pos = 10.^(-k);
[x_pos f_taylor_pos(x_pos) f_taylor_neg(x_pos)]

x_neg = -10.^(-k);
[x_neg f_taylor_pos(x_neg) f_taylor_neg(x_neg)]
```

No bad behaviour is observed since the 1 in both formulas dominates the sum and numerical cancellation does not occur.Therefore both expression can be accepted.Instead, if the sum were not dominated by the term 1 (or if you didn't notice that),the following combination could have been proposed

$$e^{-x} \approx \begin{cases} \frac{1}{1+x+\frac{1}{2}x^2+\frac{1}{6}x^3+\cdots}, & x > 0; \\ 1 - x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \cdots, & x \leq 0. \end{cases}$$

The choice of the previous expression for $x > 0$ should not display anynumerical cancellation because all the terms in the sum are positive.Similarly, the choice for $x < 0$ grants that the terms $-x^{2k+1}$ arepositive, again avoiding any numerical cancellation.

Homework 1.5. Consider the following integral

$$I_n(\alpha) = \int_0^1 \frac{x^n}{x + \alpha}\,\mathrm{d}x, \qquad \forall n \in \mathbb{N}, \alpha > 0.$$

a. Give an upper bound for $I_n(\alpha)$, $\forall n \in \mathbb{N}, \alpha > 0$.

b. Prove the following recursive relation between $I_n(\alpha)$ and $I_{n-1}(\alpha)$:

$$\begin{cases} I_n(\alpha) = -\alpha I_{n-1}(\alpha) + \frac{1}{n} \\ I_0(\alpha) = \ln\left(\frac{\alpha+1}{\alpha}\right) \end{cases}$$

c. Employing the previous relation, compute $I_{40}(\alpha = 8)$ and comment the obtained results.

d. Write a numerically stable recursive relation for $I_{40}(\alpha = 8)$.

Solution Homework 1.5.

a. Give an upper bound for $I_n(\alpha)$, $\forall n \in \mathbb{N}, \alpha > 0$.

$$\int_0^1 \frac{x^n}{x+\alpha}\,\mathrm{d}x \leq \int_0^1 \frac{x^n}{\alpha}\,\mathrm{d}x = \frac{1}{\alpha}\frac{1}{n+1}$$

b. Prove the following recursive relation between $I_n(\alpha)$ and $I_{n-1}(\alpha)$:

$$\begin{cases} I_n(\alpha) = -\alpha I_{n-1}(\alpha) + \frac{1}{n} \\ I_0(\alpha) = \ln\left(\frac{\alpha+1}{\alpha}\right) \end{cases}$$

It's a consequence of the following:

$$I_n(\alpha) = \int_0^1 \frac{x^n}{x+\alpha}\,\mathrm{d}x = \int_0^1 \frac{x^n + \alpha x^{n-1}}{x+\alpha} - \alpha \int_0^1 \frac{x^{n-1}}{x+\alpha}\,\mathrm{d}x = +\frac{1}{n} - \alpha I_{n-1}(\alpha)$$

c. Employing the previous relation, compute $I_{40}(\alpha = 8)$ and comment the obtained results.

```
clear all;
format short e
n = 40;
% alpha = 1/8 not requested in the homework
for (alpha = [1/8, 8])
   I(1) = log((alpha+1)/alpha);
   for (k = 1:n)
      I(k+1) = -alpha*I(k) + 1/k;
   end
   recursion_integral = I(n+1);
   upper_bound = (1/alpha)*(1/(n+1));
   exact_integral = quadl(@(x) (x.^n)./(x+alpha), 0, 1, 1e-16); % Exact value (not requested)
   [recursion_integral, upper_bound, exact_integral]
end
```

The recursive approximation of $I_{40}(\alpha = 8)$ is $1.6389 \cdot 10^{18}$: this result is for sure incorrect because it violates the upper bound for $I_{40}(\alpha = 8)$.This is due to the finite precision we use and the error propagation: the initial value $I_0(\alpha) = \ln\left(\frac{\alpha+1}{\alpha}\right)$ is represented with a certain error ε; denoting by $fl(y)$ the floating point representation of the number y, we have

$$\begin{aligned} fl(I_0(\alpha)) &= I_0(\alpha) + \varepsilon, \\ fl(I_1(\alpha)) &= -\alpha fl(I_0(\alpha)) + 1 = -\alpha(I_0(\alpha) + \varepsilon) + 1 = I_1(\alpha) - \alpha\varepsilon, \\ &\dots \\ fl(I_k(\alpha)) &= I_n(\alpha) + (-1)^k \alpha^k \varepsilon, \end{aligned}$$

Therefore, at the final (n-th) step the error is multiplied by a factor α^n, which result in an amplification of the error if $\alpha > 1$. Instead, for $\alpha < 1$, the error is damped and the recursive relation is stable (e.g. see the previous test for $\alpha = 1/8$).

d. Write a numerically stable recursive relation for $I_{40}(\alpha = 8)$.

The idea is to transform the factor α on the RHS of the recursive relation in a factor $\frac{1}{\alpha}$; this can be achieved inverting the recursive relation:

$$\begin{cases} I_{k-1}(\alpha) = -\frac{1}{\alpha} I_k(\alpha) + \frac{1}{k\alpha} \\ lim_{k \to +\infty} I_k(\alpha) = 0 \end{cases}$$

The final value $lim_{k\to+\infty} I_k(\alpha)$ is equal to zero because $0 \leq I_k(\alpha) \leq \frac{1}{\alpha}\frac{1}{k+1} \to 0$, as $k \to +\infty$.

```
clear all;
n = 40;
big = 1000;
% alpha = 1/8 not requested in the homework
for (alpha = [1/8, 8])
   I(big + 1) = 0;
   for (k = big:-1:n+1)
      I(k) = -1/alpha*I(k+1) + 1/(k*alpha);
   end
   recursion_integral = I(n+1);
   upper_bound = (1/alpha)*(1/(n+1));
   exact_integral = quadl(@(x) (x.^n)./(x+alpha), 0, 1, 1e-16); % Exact value (not requested)
   [recursion_integral, upper_bound, exact_integral]
end
```

The recursion is now stable for $\alpha > 1$ (and note that, now, an additional error is present, because the final condition can be set for an arbitrary large $k = \overline{k}$ insteady of $k = +\infty$).

Homework 1.6. Given the following sequence:

$$\begin{cases} x_{n+1} = 2^{n+1} \left[\sqrt{1 + \dfrac{x_n}{2^n}} - 1 \right] \\ x_0 > -1 \end{cases}$$

for which $\lim_{n \to +\infty} x_n = \ln(1 + x_0)$.

a. Set $x_0 = 1$, compute $x_1, \ x_2, \ \ldots, \ x_{71}$ and explain the obtained results.

b. Transform the sequence in an equivalent one that converges to the theoretical limit.

Solution Homework 1.6.

a. Set $x_0 = 1$, compute $x_1, \ x_2, \ \ldots, \ x_{71}$ and explain the obtained results.

```
clear all, close all, clc
format long

n_max = 71;
x = zeros(n_max+1, 1);

x(1) = 1;     % x_0 set to 1
for n = 0:n_max-1
  x(n+2) = 2^(n+1)*(sqrt(1 + x(n+1)/2^n) - 1); % Pay attention to the indexing!!!
end
x(end)
x_lim = log(1 + x(1))

figure
hold on, box on
plot([0:71], x, 'bx-','Linewidth',3)
plot([0:71], x_lim*ones(n_max+1,1)', 'r-','Linewidth',3)
axis([-1 72 -0.05 1.05])
set(gca,'LineWidth',2)
set(gca,'FontSize',16)
```

When computing the sequence with MATLAB we find that x_{71} differs a lot from $\ln(1 + x_0) = \ln(2)$. The plots also shows that $x_n = 0$, $\forall n \geq n^* = 52$: this is due to numerical cancellation effects, because at $n = n^*$ it holds $\frac{x_{n^*}}{2^{n^*}} <$ `eps`, hence in finite precision arithmetic $x_{n^*} = 0$!

```
x(53)
```

The value of n^* can also be computed with the following argument: since (hopefully) numerical cancellation will occur for large n

$$x_n \approx \ln(1 + x_0)$$

therefore

$$\frac{x_n}{2^n} < \texttt{eps} \text{ is approximately equivalent to } \frac{\ln(1 + x_0)}{2^n} < \texttt{eps}$$

which can be solved for n, finding

$$n > \log_2\left(\frac{\ln(1 + x_0)}{\texttt{eps}}\right)$$

```
x_0 = 1;
n = log(log(1 + x_0)/eps)/log(2)
```

b. Transform the sequence in an equivalent one that converges to the theoretical limit.After a razionalization, the recurrence can be written as

$$x_{n+1} = 2^{n+1}\left[\sqrt{1 + \frac{x_n}{2^n}} - 1\right] = \frac{2x_n}{\sqrt{1 + \frac{x_n}{2^n}} + 1}$$

```
clc, clear all

n_max = 71;
x = zeros(n_max+1, 1);

x(1) = 1;      % x_0 set to 1
for n = 0:n_max-1
  x(n+2) = 2*x(n+1)/(sqrt(1 + x(n+1)/2^n) + 1);
end
x(end)

% The error in this case is
x(end) - log(1 + x(1))

figure
hold on, box on
plot([0:n_max], x,'bx-','Linewidth',3)
plot([0:n_max], log(1+x(1))*ones(n_max+1), 'r-','Linewidth',3)
axis([-1 72 0.67 1.02])
set(gca,'LineWidth',2)
set(gca,'FontSize',14)

% Using the previous calculations we know that (1 + xn/2^n) approximately 1 for n >= 52, so x_
(n+1) = 2*x_n/(1 + 1) = x_n
```